

## EXERCISE 02 - AGGREGATES AND OPERATIONS

DT MBENGA

DATA ANALYTICS

BRIGHTLEAN

4/09/2026

Q1

▶ Run all (1000) ✓ 12:05 PM (11s) workspace.default

1 /\* Exercise 2 - Question 1

2 List all distinct departments

3 \*/

4

5 SELECT DISTINCT department

6 FROM students;

Add parameter

Table +

|   | <sup>B</sup> <sub>A C</sub> department |
|---|----------------------------------------|
| 1 | IT                                     |
| 2 | HR                                     |
| 3 | Finance                                |

Q2

SQL Editor

Exercise 2 - Question 2

Run all (1000)

✓ 12:39 PM (12s)

workspace.default

1

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7

/\* Exercise 2 - Question 2

List all distinct departments

\*/

SELECT

department, avg(age)

FROM students

GROUP BY department

Add parameter

Table

+

>

|   | department | 1.2 avg(age) |
|---|------------|--------------|
| 1 | IT         | 20.5         |
| 2 | HR         | 22           |
| 3 | Finance    | 23           |

Q3

SQL Editor

Exercise 2 - TABLES

Exercise 2 - Question

Run all (1000)

✓ 12:42 PM (1s)

workspace.default

1

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/\* Exercise 2 - Question 3

Show departments with more than 1 student

\*/

SELECT department, COUNT(\*) AS student\_count

FROM students

Add parameter

Table

+

|   | A <sup>B</sup> <sub>C</sub> department | 1 <sup>2</sup> <sub>3</sub> student_count |
|---|----------------------------------------|-------------------------------------------|
| 1 | IT                                     | 2                                         |
| 2 | HR                                     | 2                                         |



Q4

SQL Editor

Exercise 2 - TABLES

Exercise 2 - Question 3

Exercise 2 - Question 4

Run all (1000)

12:43 PM (1s)

workspace.

default

1

2

3

4

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6

/\* Exercise 2 - Question 4

Students aged between 21 and 23

\*/

SELECT student\_id, name, age, department

FROM students

Add parameter

Table

+

123 student\_id

ABC name

123 age

ABC department

1

2

3

4

2

3

4

5

Bob

Charlie

Diana

Eve

22

21

23

22

HR

IT

Finance

HR

Q5

▶ Run all (1000) ✓ 12:51 PM (1s) workspace.default

1

2

3

4

5

6

/\* Exercise 2 - Question 5

IT or HR students older than 21

\*/

SELECT student\_id, name, age, department

FROM students

Add parameter

Table +

|   | student_id | name | age | department |
|---|------------|------|-----|------------|
| 1 | 2          | Bob  | 22  | HR         |
| 2 | 5          | Eve  | 22  | HR         |

Q6

▶ Run all (1000) ✓ 12:52 PM (4s) workspace.default

15

16

17

18

19

20

```
/* Exercise 2 - Question 6 */
SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;
```

Add parameter

← Results 4 of 4 →

Table

+

|   | <sup>A</sup> <sub>C</sub> department | <sup>1</sup> <sub>3</sub> total_credits |
|---|--------------------------------------|-----------------------------------------|
| 1 | IT                                   | 11                                      |

>

Q7

SQL Editor

Exercise 2 - TABLES

Exercise 2 - Question 7

Exercise 2 - Question 7

Run all (1000)

12:54 PM (<1s)

workspace.

default

```
1  /* Exercise 2 - Question 7 */
2
3  SELECT course_id, course_name, department, credits
4  FROM courses
5  WHERE credits != 4;
```

Add parameter

Table

+

|   | 1 <sup>2</sup> <sub>3</sub> course_id | A <sup>B</sup> <sub>C</sub> course_name | A <sup>B</sup> <sub>C</sub> department | 1 <sup>2</sup> <sub>3</sub> credits |
|---|---------------------------------------|-----------------------------------------|----------------------------------------|-------------------------------------|
| 1 | 101                                   | SQL Basics                              | IT                                     | 3                                   |
| 2 | 104                                   | Excel                                   | Finance                                | 2                                   |
| 3 | 105                                   | Statistics                              | HR                                     | 3                                   |

Q8

▶ Run all (1000) ✓ 12:55 PM (1s) workspace.default

```
1  /* Exercise 2 - Question 8 */
2
3  SELECT course_id, course_name, credits
4  FROM courses
5  ORDER BY credits DESC
6  LIMIT 3;
```

Add parameter

Table +

|   | course_id | course_name  | credits |
|---|-----------|--------------|---------|
| 1 | 102       | Python       | 4       |
| 2 | 103       | Data Science | 4       |
| 3 | 101       | SQL Basics   | 3       |

Q9

SQL Editor

Exercise 2 - TABLES

Exercise 2 - Question 9

Exercise

Run all (1000) ✓ 12:57 PM (4s) workspace. default

```
13 (3, 3, 103, 90),
14 (4, 4, 104, 88),
15 (5, 5, 105, 82);
16
17 /* Exercise 2 - Question 9 */
18 SELECT
19     MAX(grade) AS max_grade,
20     MIN(grade) AS min_grade,
21     AVG(grade) AS avg_grade
22 FROM enrollments;
```

Add parameter

Results 4 of 4

Table

|   | $1^2_3$ max_grade | $1^2_3$ min_grade | 1.2 avg_grade |
|---|-------------------|-------------------|---------------|
| 1 | 90                | 78                | 84.6          |

Run all (1000) ✓ 12:58 PM (<1s) workspace.default

1

2

3

4

5

/\* Exercise 2 - Question 10 \*/

SELECT course\_id, COUNT(\*) AS enrollment\_count

FROM enrollments

GROUP BY course\_id;

Add parameter

Table +

>

|   | course_id | enrollment_count |
|---|-----------|------------------|
| 1 | 101       | 1                |
| 2 | 102       | 1                |
| 3 | 103       | 1                |
| 4 | 104       | 1                |
| 5 | 105       | 1                |

Q11



▶ Run all (1000) ▾



01:07 PM (4s)



workspace. default ▾



```
16  (5, BULCH, HR, 54000, 3500);
```

```
17
```

```
18  /* Exercise 2 - Question 11 */
```

```
19
```

```
20  SELECT
```

```
21      department,
```

```
22      SUM(salary) AS total_salary,
```

```
23      SUM(bonus) AS total_bonus
```

```
24  FROM salaries
```

```
25  GROUP BY department;
```

Add parameter



Results 4 of 4



Table ▾



|   | <sup>A</sup> <sub>C</sub> department | <sup>1</sup> <sub>2</sub> <sup>3</sup> total_salary | <sup>1</sup> <sub>2</sub> <sup>3</sup> total_bonus |
|---|--------------------------------------|-----------------------------------------------------|----------------------------------------------------|
| 1 | IT                                   | 122000                                              | 10500                                              |
| 2 | HR                                   | 109000                                              | 7500                                               |
| 3 | Finance                              | 70000                                               | 6000                                               |

Q12

The screenshot shows a SQL Editor window with a tab titled "Exercise 2 - Question 12". The editor contains the following SQL query:

```
1  /* Exercise 2 - Question 12 */  
2  SELECT  
3      department,  
4      AVG(salary) AS avg_salary  
5  FROM salaries  
6  GROUP BY department  
7  HAVING AVG(salary) > 55000;
```

Below the query editor, there is a button labeled "Add parameter". Below that, a table is displayed with the results of the query. The table has two columns: "department" and "avg\_salary". The results are as follows:

|   | department | avg_salary |
|---|------------|------------|
| 1 | IT         | 61000      |
| 2 | Finance    | 70000      |

Q13

SQL Editor

Exercise 2 - Question 13Exercise 2 - Question 14Exercise 2 - Question 15

Run all (1000)01:09 PM (<1s)workspace.default

1

2

3

4

5

6

7

8

/\* Exercise 2 - Question 13 \*/

SELECT

employee\_id,

name,

salary,

bonus,

salary + bonus AS total\_compensation

Add parameter

Table

employee\_id

name

salary

bonus

total\_compensation

1

1

Tom

60000

5000

65000

2

3

Spike

70000

6000

76000

3

4

Tyke

62000

5500

67500

Q14

SQL Editor Exercise 2 - Question 13 Exercise 2 - Question 14 ×

Run all (1000) ✓ 01:11 PM (5s) workspace default

```
14 (4, 'Website', 'Marketing', 80000),
15 (5, 'HR Portal', 'HR', 50000);
16
17
18 /* Exercise 2 - Question 14 */
19 SELECT
20     department,
21     SUM(budget) AS total_budget,
22     AVG(budget) AS avg_budget
23 FROM projects
24 GROUP BY department
25 HAVING AVG(budget) > 70000;
```

> Add parameter

← Results 4 of 4 → Table +

|   | <sup>A</sup> <sub>C</sub> department | <sup>1</sup> <sub>3</sub> total_budget | <sup>1.2</sup> avg_budget |
|---|--------------------------------------|----------------------------------------|---------------------------|
| 1 | IT                                   | 270000                                 | 135000                    |
| 2 | Finance                              | 80000                                  | 80000                     |

Q15

Run all (1000) ✓ 01:12 PM (1s) workspace: default

1

2

3

4

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9

10

/\* Exercise 2 - Question 15 \*/

SELECT

project\_id,

project\_name,

department,

budget

FROM projects

WHERE budget BETWEEN 50000 AND 120000

AND department != 'Marketing';

Add parameter

> Table +

|   | 1 <sup>2</sup> <sub>3</sub> project_id | A <sup>B</sup> <sub>C</sub> project_name | A <sup>B</sup> <sub>C</sub> department | 1 <sup>2</sup> <sub>3</sub> budget |
|---|----------------------------------------|------------------------------------------|----------------------------------------|------------------------------------|
| 1 | 1                                      | AI App                                   | IT                                     | 120000                             |
| 2 | 2                                      | Payroll System                           | Finance                                | 80000                              |
| 3 | 5                                      | HR Portal                                | HR                                     | 50000                              |